

Electronics (1)

電子學(一)

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Documentation Outlines

- **Integrated Device-to-System Progression:** The material moves from basic semiconductor physics and diode models to complex multi-transistor amplifiers and feedback systems.
- **Strategic Sequencing:** The curriculum identifies "essential" versus "optional" topics, with a specific focus on bipolar and MOS devices as foundational elements for subsequent work.
- **Analytical Primacy over Simulation:** Simulation tools (SPICE) are introduced only after students have established a baseline of analytical mastery and fundamental understanding.
- **Rigorous Assessment:** Evaluation methods utilize "twisted" versions of textbook problems to ensure students solve all provided exercises rather than relying on rote memorization.

Course Structure

Period	Focus Areas	Chapters Covered
Q1	Intro to Microelectronics, Semiconductor Physics, Diode Models, Bipolar Transistors	Chapters 1~4
Q2	Bipolar Amplifiers, MOS Devices, CMOS Amplifiers, Op Amps (Black Box)	Chapters 5~8
Q3	Cascodes and Current Mirrors, Differential Pairs	Chapters 9~10
Q4	Frequency Response, Feedback and Stability	Chapters 11~12

Core Thematic Analysis

- **Device Foundations and Nonlinearity :**
- The curriculum begins by establishing the "big picture" of analog and digital signals before diving into the physics of devices.
 1. **Semiconductor Physics (Chapter 2):** Covers electrons/holes, doping, drift/diffusion, and the *pn* junction. The text uses a deliberately slow pace and multiple angles to ensure student comprehension of device physics before constructing circuit models.
 2. **Diode Models (Chapter 3):** Serves to familiarize students with the *pn* junction as a nonlinear device. It introduces the concept of linearizing nonlinear models to simplify analysis and covers essential circuits like rectifiers and limiters.

Core Thematic Analysis

- **Transistor and Amplifier Mastery :**
- The transition to transistors (Bipolar and MOS) is handled through a parallel structure, introducing them as voltage-controlled current sources.
 1. **Bipolar Amplifiers (Chapter 5):** Cited as the longest and most critical chapter, it uses a bottom-up approach to establish input/output impedances, biasing, and small-signal analysis. It reinforces synthesis and circuit topologies through "What if?" examples.
 2. **MOS Devices and CMOS Amplifiers (Chapters 6-7):** Parallels the bipolar chapters. Due to time constraints, complex phenomena like the body effect and velocity saturation are introduced briefly but neglected in later chapters to maintain focus.
 3. **Operational Amplifiers (Chapter 8):** Treated as a "black box." The curriculum suggests teaching this after amplifier topologies so students understand internal gain limitations and the relevance of op amps compared to differential amplifiers.

Core Thematic Analysis

- Advanced Integrated Circuit Concepts
- The second phase of the course shifts toward integrated circuit design.
 1. **Cascodes and Current Mirrors (Chapter 9):** Provides the necessary background for constructing differential pairs with active loads.
 2. **Differential Amplifiers (Chapter 10):** Explores large-signal and small-signal behavior, bridging the gap between previous amplifier studies.
 3. **Feedback and Stability (Chapter 12):** Acknowledged as the most difficult topic for undergraduates. The curriculum employs a step-by-step procedure to analyze feedback circuits, particularly regarding loading effects and stability conditions (Phase Margin, Frequency Compensation).

Core Thematic Analysis

- Specialized and Semester-Based Topics
- Chapters 13 through 17 are intended for semester-based systems or elective depth:
 1. **Oscillators (Chapter 13):** Covers ring, LC, phase shift, Wien-bridge, and crystal oscillators.
 2. **Output Stages (Chapter 14):** Analyzes high-power delivery circuits, including push-pull stages and heat dissipation.
 3. **Digital CMOS (Chapter 16):** Provides a foundation for subsequent digital courses, focusing on CMOS inverters and gates while excluding TTL and ECL due to time constraints.

Assessment and Simulation Tools

- SPICE Integration
- Simulation is positioned as a secondary, albeit vital, tool.
 1. **Timing:** Introduced late in the first course (around Chapter 5).
 2. **Purpose:** The objective is to master a subset of SPICE commands for circuit simulation.
 3. **Prerequisite:** Students must first develop analytical skills. The documentation argues that students appreciate simulation's utility more when dealing with circuits containing a large number of devices.